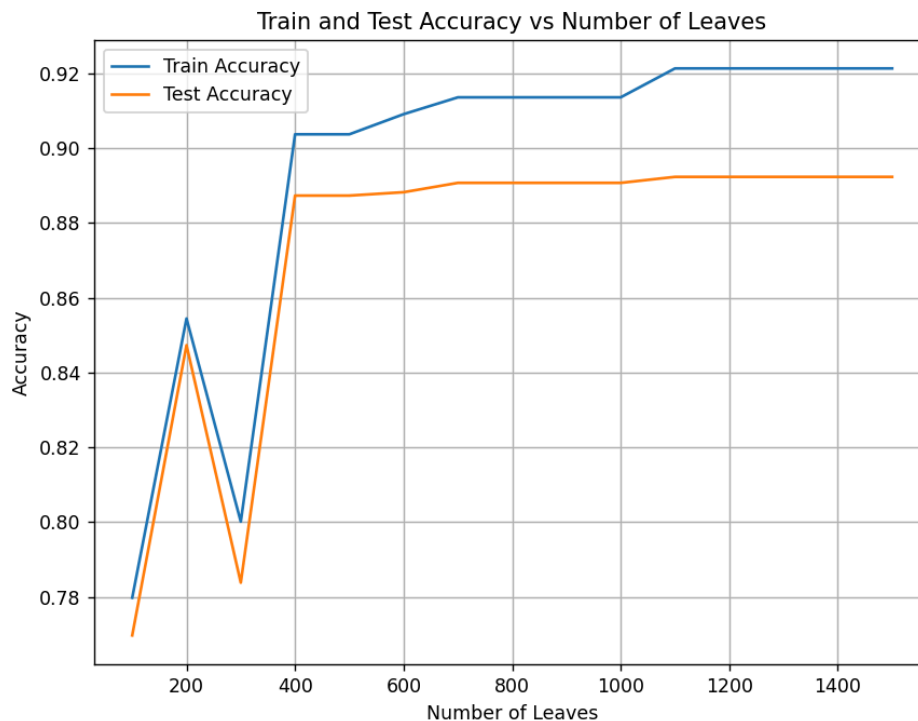
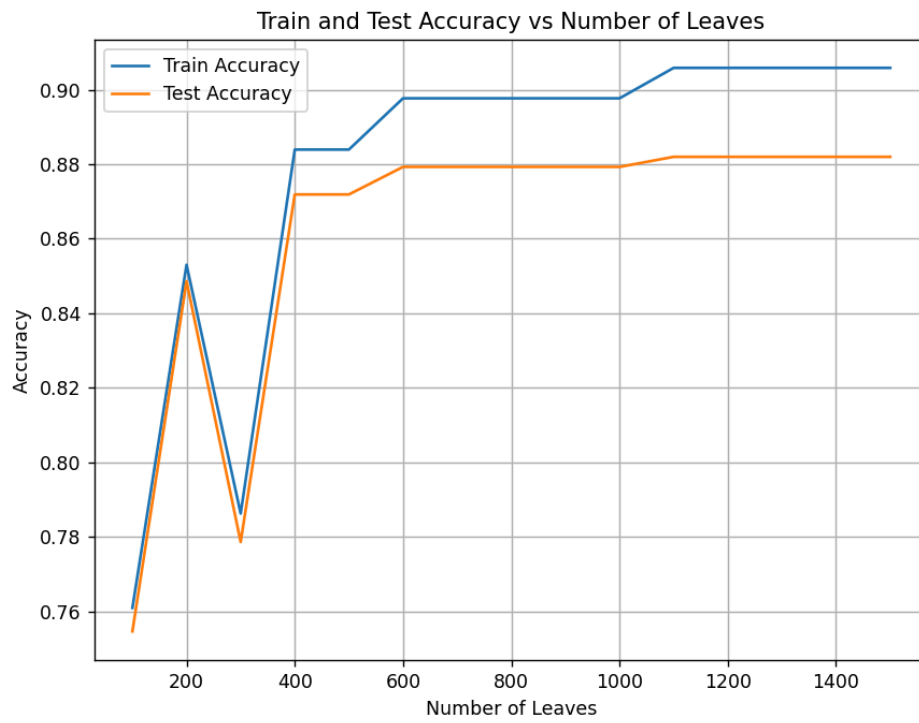


Stepsize = 500



1. As the number of leaves increases, the train and test accuracy improve steadily with a slight dip at the beginning. A larger tree can fit the training data more closely, leading to lower training error. However, after a certain number of leaves (around 600–800 leaves in this case), test accuracy plateaus around 89%, while train accuracy continues to rise above 91%. This divergence between train and test curves indicates overfitting. The model is starting to memorize training data patterns that don't generalize well to unseen data. Beyond that point, adding more leaves increases model complexity but doesn't provide generalization benefits.
2. The step size controls how many candidate thresholds are considered per feature when finding the best split. A large step size like 1000 can cause the algorithm to skip a lot of possible thresholds, making splits coarse and potentially suboptimal. Training is a bit faster with larger step size, but accuracy drops slightly because the tree can't find fine-grained decision boundaries. A very small step size like 1 will cause the algorithm to check every possible threshold for each feature, which can yield better accuracy, since the best split is found precisely. However, this comes at the cost of much higher computation time.

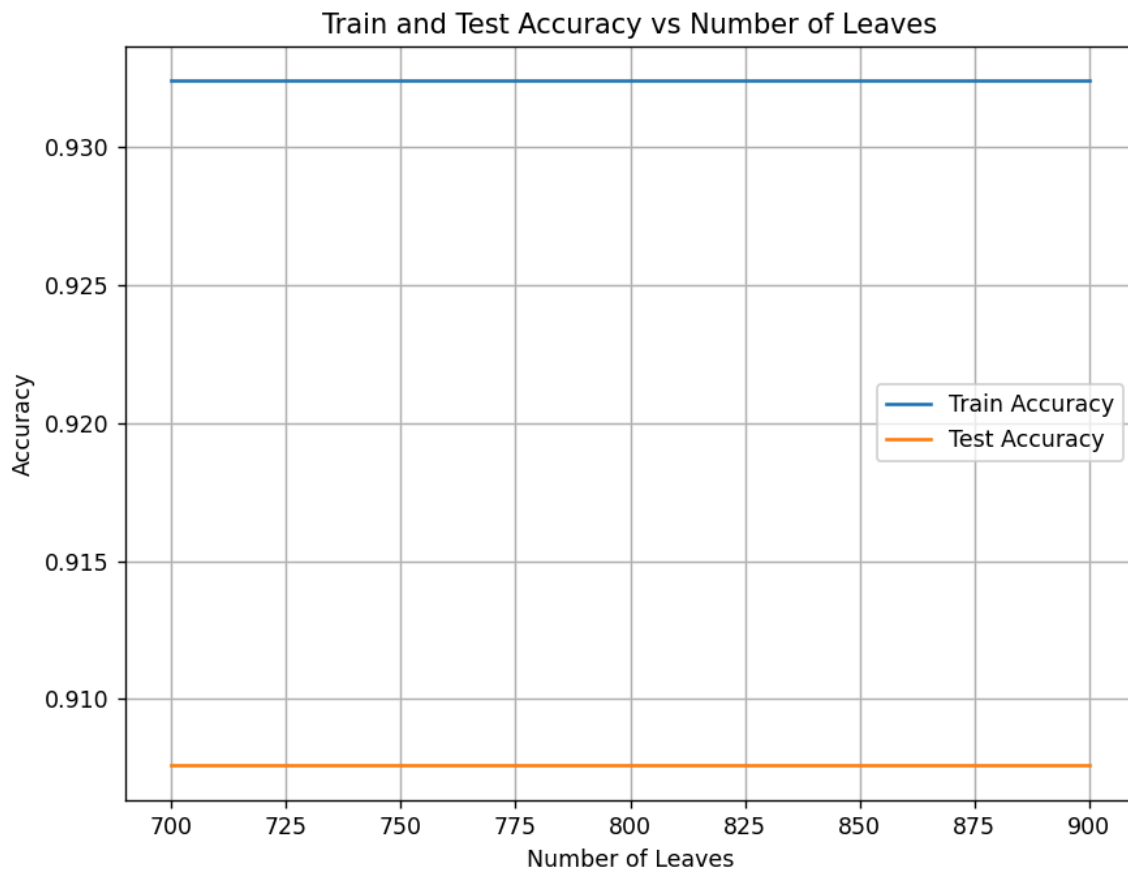
Stepsize = 1000



Stepsize = 250



Stepsize = 50 (tested on fewer prime examples to get above 90%)



Test accuracy array: [np.float64(0.9076), np.float64(0.9076), np.float64(0.9076)]